



Full length article

DFMKE: A dual fusion multi-modal knowledge graph embedding framework for entity alignment

Jia Zhu^a, Changqin Huang^{a,*}, Pasquale De Meo^b^a Zhejiang Normal University, China^b University of Messina, Italy

ARTICLE INFO

Keywords:

Knowledge graph

Entity alignment

Neural networks

Multi-modal knowledge

ABSTRACT

Entity alignment is critical for multiple knowledge graphs (KGs) integration. Although researchers have made significant efforts to explore the relational embeddings between different KGs, existing approaches may not describe multi-modal knowledge well in some tasks, e.g., entity alignment. In this paper, we propose DFMKE, a dual fusion multi-modal knowledge graph embedding framework, to address entity alignment. We first devise an early fusion method for fusing features of multi-modal entity representations of a KG. Simultaneously, multiple representations of various types of knowledge are generated independently by various techniques and fused by a low-rank multi-modal late fusion method. Finally, the outputs of early and late fusion methods are combined using a dual fusion scheme. DFMKE provides an ultimate fusion solution by leveraging the advantages of early and late fusion methods. Extensive experiments on two public datasets show that the DFMKE outperforms state-of-the-art methods by a significant margin achieving at least 10% more regard to Hits@n and MRR metrics.

1. Introduction

Entity alignment is critical for integrating distinct knowledge graphs (KGs) by connecting entities that refer to the same real-world object. Because most KGs are created with a specific purpose, different KGs showcase different representations for the same concepts [1].

Early studies on entity alignment primarily focus on attribute similarity [2,3], which frequently suffers from attribute heterogeneity, making entity alignment error-prone [4]. Subsequently, other approaches to aligning entities in KGs required human intervention [5], or extra-resources [6], that is, a description of involved entities and their relations. The methods above require many examples to work in downstream tasks correctly, but obtaining a sufficiently large number of examples in real-life applications is challenging and expensive. More recently, some authors suggested semi-supervised models perform entity alignment in KGs; that is, they sought new algorithmic approaches which benefit from both labeled and unlabeled entities [7,8].

For instance, [7] proposed the SEA (Semi-supervised Entity Alignment) framework. The SEA framework extends the popular TransE embedding method [9] to deal with the degree difference of entities. SEA, in particular, applies adversarial training to prevent entities with a similar degree of popularity are aggregated in the same region of the embedding space during the training step. After generating the

embedding of two input KGs, SEA constructs and optimizes a suitable objective function that contains both labeled and unlabeled entities.

A further exciting approach is KECG (Knowledge Embedding model and Cross-Graph model), proposed in [8]. The KECG approach formulates the problem of aligning entities as an optimization problem in which the objective function is the sum of two components. The first component of the objective function to optimize is related to the so-called cross-graph model, which captures inner KG structures and alignments between entities in different KGs. The cross-graph model applies an attention mechanism, and, in particular, it is built utilizing an extended GAT as an encoder: in this way, the KECG system can ignore unimportant nodes. We observe that KECG uses both labeled and unlabeled examples in the cross-graph model construction process; thus, it has to be regarded as a semi-supervised approach. The second component of the objective function to optimize uses the TransE algorithm to learn the embedding of entities and relations in different KGs, and then it aligns these representations into a unified vector space.

SEA and KECG have achieved excellent performance on a range of popular datasets, but they cannot handle multi-modal KGs: in other words, KGs are often populated by heterogeneous data such as texts, numbers, or images to describe the same piece of reality. Distinct knowledge forms play a crucial role as auxiliary data to complete KGs and perform entity alignment; thus, we are strongly motivated to design

* Corresponding author.

E-mail addresses: jjazhu@zjnu.edu.cn (J. Zhu), cqhuang@zju.edu.cn (C. Huang).

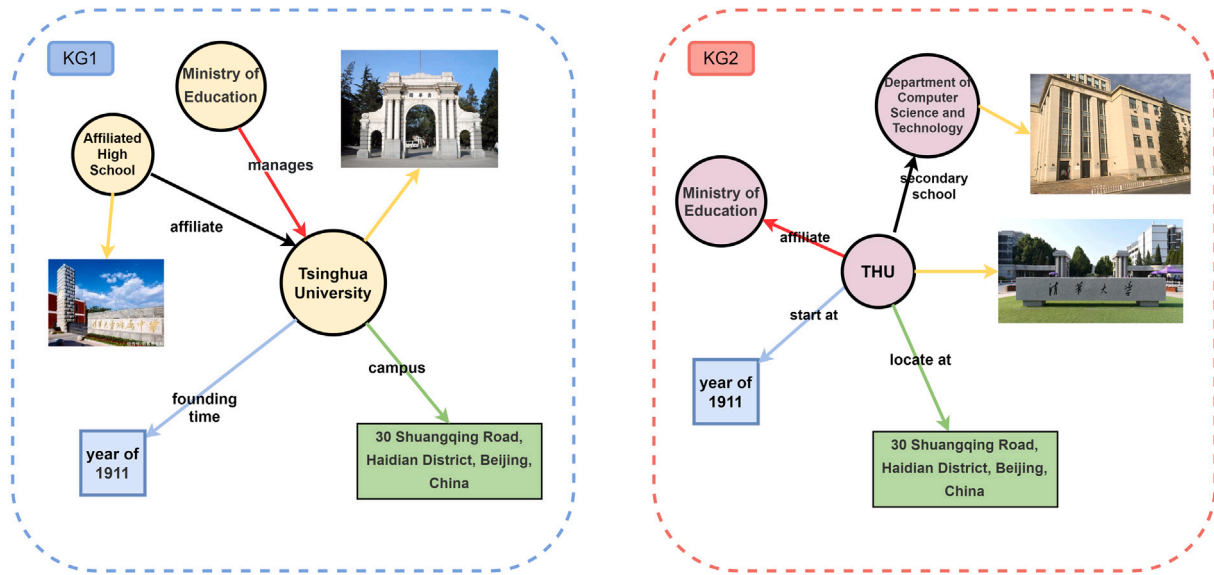


Fig. 1. Illustration of entity alignment between multi-modal knowledge graphs.

new entity matching algorithms that can exploit their full potential multi-modal data.

Fig. 1 provides an example of entity alignment for multi-modal KGs, and it clarifies the peculiarities arising from the presence of multi-modal data in a KG. Here, the images associated with the entity “THU” indicate that the type of this entity is “university” in this case. However, leveraging multi-modal knowledge to perform entity alignment is not trivial. The inevitable heterogeneity among different modalities makes the entity alignment task challenging. For example, in Fig. 1, it is difficult to conclude that the “Tsinghua University” entity in KG1 and the “THU” entity in KG2 refer to the same object using only images or text information.

Recent studies [10–12] have put forth several models that combine multi-modal data from KGs into a joint embedding and enable the alignment model to modify modality weights automatically. However, the approaches above did not consider modal correlation at the feature level, and thus, these approaches may achieve poor results if multiple modalities are highly correlated. In addition, most existing works do not work well when seed entities, a list of labeled entities across KGs to initialize the training process, are not broadly available.

To address the issues above, we propose a dual fusion multi-modal knowledge graph embedding framework (DFMKE) for modeling the entity associations of multi-modal KGs and locating entities referring to the same real-world identity. Specifically, we propose an early fusion strategy to perform feature fusion among different modalities, which can exploit correlations between low-level features of each modality. Then, we discriminatively generate knowledge representations for each modality and design a late fusion method based on low-rank weight decomposition to leverage knowledge from multiple modalities for the entity alignment task. This work offers three contributions:

- To alleviate the inconsistency of original data in each modality, we propose a dual multi-modal knowledge graph embedding framework called DFMKE that can incorporate the advantages of both early fusion and late fusion techniques for entity alignment with joint training. The main idea of DFMKE is to integrate knowledge representations of multiple modalities from separate spaces to a shared spaces.
- We present a novel late fusion method for multi-modal fusion using modality-specific low-rank factors. This method can easily combine the output features from the early fusion method and reduce the computational complexity caused by input transformation into a tensor.

- We prove the performance of the DFMKE by conducting experiments on two public multi-modal datasets, and we compared DFMKE with several state-of-the-art entity alignment methods. DFMKE works well even when no seed entities are available to initialize the training process. We also offer interpretable analysis in our work by conducting ablation studies on early and late fusion modules’ contributions.

The remainder of the paper is organized as follows. Section 2 briefly discusses the main existing works; Section 3 introduces the technical details of the work; Section 4 reports and comments on the experimental results according to various benchmarks; Section 5 concludes this paper.

2. Related literature

KGs are a powerful tool to efficiently organize, manage and retrieve a large body of information which is usually represented as a collection of RDF triplets of the form (head, predicate, tail) [13].

Two core problems in the KG research area are: (a) Link Prediction, i.e., we aim at completing triplets of the form (head?, predicate, tail) (in which the head is not specified) or (head, predicate, tail?) (in which the tail is not specified); (b) Entity Matching, i.e., given two KGs, we wish to find pair of entities (one from each KG) which correspond to the same entity (and, thus, entities are related by a sameAs predicate).

Our work mainly falls in two areas, entity alignment and multi-modal knowledge graphs; in what follows, we describe the main approaches related to ours.

2.1. Entity alignment in knowledge graphs

Entity alignment (also known as entity matching or entity resolution) refers to the task of identifying pairs of entities from different KGs that refer to the same object in the real world.

Early approaches to dealing with the entity alignment problem in KGs use discriminative features such as names, descriptive annotations, and relational structures in each KGs [3,6,14–16]. For instance, Raimond et al. [15] proposed a method called LD-Mapper to utilize entity nearest neighbor similarity and string similarity. Later, it was suggested to use SILK to calculate string equality and similarity, numeric similarity, and other metrics to determine how similar entities are. The linguistic and schematic heterogeneity between independently created KGs was the main challenge the systems above needed to deal with.

More recently, embedding-based methods have been considered an effective approach to handling the entity alignment problem [17–22]. Embedding-based methods map KGs onto vectors of a low-dimensional space so that the embedding space's geometrical structure can reveal semantic relatedness between pairs of entities [9]. For example, Zhu et al. [23] proposed IPTransE, an iterative method through joint knowledge embedding. Li et al. [24] used the unaligned entities to improve performance by being aware of the degree of difference in adversarial training. Concerning the lack of aligned entities for training, Pei et al. [25] proposed a semi-supervised entity alignment method with a cycle-consistency-based translation loss to leverage the unaligned entity instead of only using labeled aligned entities. Later, Li et al. [26] proposed a semi-supervised entity alignment method called KECG that utilized an extended graph attention network for encoding heterogeneous KGs and performing entity alignment by propagating alignment information over the entire graphs. Their experiments proved that the method does not require a sufficient number of aligned entities. However, These methods have not been proven suitable for multi-modal KGs.

We can identify three main categories of approaches to performing KG embedding, namely:

1. Translational Models [9,27,28]
2. Semantic Matching Models [29–31]
3. Deep Models [32,33]

We refer to [34] for a detailed survey of existing approaches. Often, KGs contain multi-modal data (text, images, and so on). Thus, it is convenient to consider these data as a tool to improve the accuracy of an approach to finding matching entities. In the next section, we introduce related literature on multi-modal KGs.

2.2. Multi-modal knowledge graphs

Holzinger et al. [35] recently observed that the direct involvement of a human-in-the-loop can be enormously helpful when constructing a KG as an initial connector, especially because it bridges the gap to the increasingly important topic of explainability and causability. We can successfully map explainability with causability through KGs if we can develop a human–AI interfaces that allow domain experts to ask questions interactively to gain deep insight into the underlying independent factors of a result. However, to develop such a system, there are many issues we need to resolve, such as link prediction and entity matching problems. Most of the previous research work mainly uses the relational structure of a KG to address the link prediction and entity matching problems. Thus, they neglect crucial auxiliary information such as text, numerical values, and images associated with entities [36–38].

To this end, an increasing number of approaches combine the relational structure of a KG with other types of information to better manage the link prediction and entity matching problems. KGs containing various data types such as numerical attributes, texts, and images are often called multi-modal knowledge graphs [39]. One of the early examples of multi-modal KG is the VisualGenome project [36]. The main output of the VisualGenome project is a KG based on WordNet, which provides annotations of categories, attributes, and relation types for each available image. In line with the work of [36], Onoro et al. [37] described IMAGEGRAPH as a publicly available KG in which entities are associated with images. They considered a novel class of queries to pose against a KG in which images are treated as arguments of the query itself.

Xie et al. [38] suggest using rich information embodied in images in constructing a KG. Xie et al. [38] describes a method called IKRL, which consists of the following steps: (1) It applies a neural image encoder (based on the AlexNet neural network [40]) on every entity image as input. The image encoder aims to generate picture representations in

the entity space and extract useful characteristics from images. (2) It applies an attention module to learn the representation of an entity, given the image instances associated with that entity, and (3) It applies a translational-based method to learn a representation of triplets and images jointly.

A further exciting approach to dealing with multi-modal KG is the MKBE (Multimodal Knowledge Base Embeddings) system discussed by Pezeshkpour et al. [39]. The MKBE system applies a specific encoder for each type of data (for instance, bidirectional GRU to encode short texts [41] and the last hidden layer of VGG pre-trained network on Imagenet [42] coupled with by compact bilinear pooling [43] to embed images). Obtained embeddings are thus combined with learning embeddings of the entities and multi-modal data. Learned embeddings (along with suitable different neural decoders) are at the basis of the multi-modal imputation model, which can create missing multi-modal values from information in the KG.

Later, an MMKG (Multi-Modal Knowledge Graphs) dataset was proposed in [44]: such a dataset is a collection of three knowledge graphs to experimentally assess the effectiveness of link prediction and entity alignment algorithms in multi-modal knowledge graphs. In the three proposed knowledge graphs, all entities are associated with numerical and visual data; the three knowledge graphs contain matchings between entities from different KGs' sameAs relations, linking entities belonging to different knowledge graphs. Liu et al. [44] also extended the learning framework described in [45] and, thus, it can manage queries such as (head?, sameAs, tail) and (head, sameAs, tail?); such a task can be considered something in between the link prediction and the entity matching problem.

Zhang et al. [12] introduced a multi-view KG embedding framework for entity alignment. This framework learns entity embeddings from three typical KG views by combining view-specific embeddings using three different algorithms. By creating entity representations for relational, visual, and numerical knowledge and fusing these representations via a multi-modal knowledge fusion module, Chen et al. [10] suggested the MMEA approach to deal with the issue of entity alignment in a multi-modal view.

Liu et al. [11] described EVA (Entity Visual Alignment), a system that uses images as pivots to align entities from different KGs. EVA operates both in semi-supervised (i.e., it makes use of a small set of labels provided by humans) and unsupervised fashion (i.e., it creates an initial seed dictionary by using the visual resemblance of entities). Therefore, both in semi-supervised and unsupervised environments, iterative learning (IL) is used to increase the number of training pivots.

In real-world datasets, the correlation between multiple modalities can be significant; thereby, such a correlation is a precious tool for embedding a multi-modal KG. Unfortunately, the approaches cited above do not consider the correlation between different modalities at the feature level, and thus, they may be less effective in calculating embeddings.

Unlike previous literature, our approach describes a method to perform multi-modal KG embedding consisting of two main phases. We call our approach DFMKE, dual fusion multi-modal knowledge graph embedding: as we will detail below, we perform feature fusion across different modalities in the first phase (called early fusion). In this way, each modality has its representation. Thus, in the second phase (called late fusion), we consider a low-rank weight decomposition algorithm that enables us to benefit from knowledge associated with different modalities to perform entity alignment.

3. Proposed framework

In this section, we first formulate the problem and then describe the technical details of the proposed framework. A multi-modal KG can be viewed as a tuple $G = (\mathcal{E}, \mathcal{I}, \mathcal{R}, \mathcal{A})$, where $\mathcal{E}$, $\mathcal{I}$, $\mathcal{R}$, $\mathcal{A}$ denote the sets of entities, images, relations and attributes, respectively. Given a pair of entities $e_s \in \mathcal{E}_s$ from source KG G_s and $e_t \in \mathcal{E}_t$ from target KG G_t , the

task of entity alignment is to match entities describing the same object in different KGs.

To tackle the entity alignment task, we propose a framework called DFMKE to learn a function f that can estimate the similarity of e_s and e_t , and such a similarity should be high if e_s and e_t refer to the same object.

As illustrated in Fig. 2, the DFMKE consists of two critical parts: multi-modal KG embeddings and dual multi-modal knowledge fusion. In the following, we will introduce the technical details of these two steps.

3.1. Multi-modal KG embeddings

In the multi-modal KG embeddings part, we generate the knowledge representations of entities from three data modalities, including a relational triple with entity associations, visual data, and entity attributes, for instance, numerical attributes and descriptions of entities. Besides, we model the structure of KG to capture both entity and relation proximity to compute the structural similarity of G_s and G_t . The following sections will go into further detail about each knowledge embedding.

KG Structure Embedding. For KG structure embedding, we design an embedding model based on graph convolutional networks (GCNs) [46] and our previous work [47]. Compared to traditional GCN-based methods for graph embedding, we treat the KG as an undirected bipartite graph and use additional nodes to represent the relationships between two entity nodes. For example, assume a triplet (e_1, r, e_2) , we assign relation vertices r_1 and r_2 for this triplet as (e_1, r_1) and (e_2, r_2) . Each relation vertex will be assigned a unique one-hot representation.

We then use the feature vector generated by Deepwalk [48] to represent each entity vertex and the unique one-hot representation for each relation vertex as input channels of GCN on this newly constructed graph. The relation vertices indicate how many neighbors with the same relation information exist between two entity vertices.

After l convolutional layers encoding, we can obtain a representation learned from the graph, including the information of entity vertices and relation vertices. Thus, each layer of GCN can be written as a non-linear function:

$$H^{(l+1)} = f(H^{(l)}, M), \quad (1)$$

where $H^{(0)}$ is the input matrix, $H^{(L)}$ is the output matrix, L is the number of layers, and M is the adjacency matrix of KG. We then set the following propagation rule:

$$f(H^{(l)}, M) = \text{ReLU}(MH^{(l)}W^{(l)}), \quad (2)$$

where $W^{(l)}$ is a weight matrix for the l th network layer, and ReLU is the activation function. Note that the multiplication by M only sums up all attributes of all neighbor vertices, not the vertex itself. Therefore, we need to add the identity matrix I to M , and Eq. (2) is updated as follows [46]:

$$f(H^{(l)}, M) = \text{ReLU}\left(\hat{D}^{-\frac{1}{2}} \hat{M} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right), \quad (3)$$

where $\hat{M} = M + I$ and $\hat{D}$ is the diagonal vertex degree matrix of $\hat{M}$. As the KG structure embedding emb_{kg} , we use the output of the final layer.

Visual Knowledge Embedding. For image embedding, we use a pre-trained RESNET model [49] as the feature extractor for all images and use the output of the last layer to represent the image. Compared to the other classical model VGG16 [50], RESNET proposed a residual network to solve the degradation problem in a deep network. The extracted feature is processed using the equation below to obtain the image embedding emb_I :

$$emb_I = W_I \cdot \text{RESNET}(I) + b_I. \quad (4)$$

Relation Embedding. A KG's fundamental component is its relations, which connect the entities. We use the well-known TransE [9]

algorithm, which considers a relation to be a translation vector from the head to the tail of an entity. We use the following scoring function with the L_2 -norm to calculate the structural similarity for a fact (h, r, t) :

$$f_{\text{rel}}(\mathbf{h}, \mathbf{r}, \mathbf{t}) = -\|\mathbf{h}^{(2)} + \mathbf{r} - \mathbf{t}^{(2)}\|. \quad (5)$$

Then we apply the logistic loss function to learn the relation embeddings emb_R as below:

$$\mathcal{L}_{emb_R} = \sum_{(h,r,t) \in \mathcal{X}^+ \cup \mathcal{X}^-} \log(1 + \exp(\alpha f_{\text{rel}}(\mathbf{h}, \mathbf{r}, \mathbf{t}))), \quad (6)$$

where α is the label of (h, r, t) whose value is 1 or -1; the symbol $\mathcal{X}^+ = \mathcal{X}_a \cup \mathcal{X}_b$ denotes the set of positive relation facts in the source and target KGs while $\mathcal{X}^-$ denotes the set of negative samples obtained by replacing the head or tail entities of positive relation facts with random ones. The goal is to maximize the probability of the prediction being positive and the probability of the negative prediction. That is to say, each sample prediction must obtain the maximum probability, and the probability of all the samples after the prediction is multiplied to get the maximum likelihood function. This loss function can guarantee better generalization ability with higher efficiency than the original TransE design.

Attribute Embedding. According to [51], modeling attributes with deep neural networks might pollute entity representations due to noise from neighbors. Thus, because in this work we focus on attribute embedding (including numerical attributes), we employ a feed-forward network for mapping attribute features into a low-dimensional space:

$$emb_A = W_A \cdot \mathbf{A} + b_A, \quad (7)$$

where emb_A is the attribute embedding; W_A is the weight matrix and b_A is the bias vector. In this way, we can improve computing efficiency with sound results.

3.2. Dual multi-modal knowledge fusion

We introduce an efficient dual fusion scheme to incorporate the advantages of features combination and outputs of embedding models from different modalities. To describe our dual fusion scheme approach, we begin by observing that multi-modal data sources have been extensively used in Machine Learning applications: in fact, each data source (or modality) provides a different type of information about the same piece of reality and, if we were able to integrate single modalities effectively, then we could reduce the impact of noisy data. The ultimate purpose of multi-modal data fusion approaches is to design more accurate and robust-to-noise Machine Learning algorithms.

A critical problem in multi-modal data fusion is the correlation among different modalities: broadly speaking, the term correlation specifies how one input modality varies if another modality varies too. In many situations, the correlation between modalities provides essential clues to deciding how to fuse two modalities [52].

Feature correlation is, in our opinion, extremely important to address the entity alignment problem for multi-modal KGs: two entities in two independent multi-modal KGs could be provided with a small amount of data, each encoded in a particular form (e.g., texts in the first KG and images in the second KG). If we could capture correlations among features, we could cleverly perform the fusion task, and thus, we could be more confident in deciding if the two entities represent the same thing.

The above observations motivate us to design a fusion scheme that can capture correlations between modalities. We recall that the fusion method can be roughly classified into early and late fusion, and we recall that early fusion methods are particularly appealing in capturing correlations between modalities; due to these reasons, our fusion scheme combines an early with a late fusion method to achieve the best of the two methods.

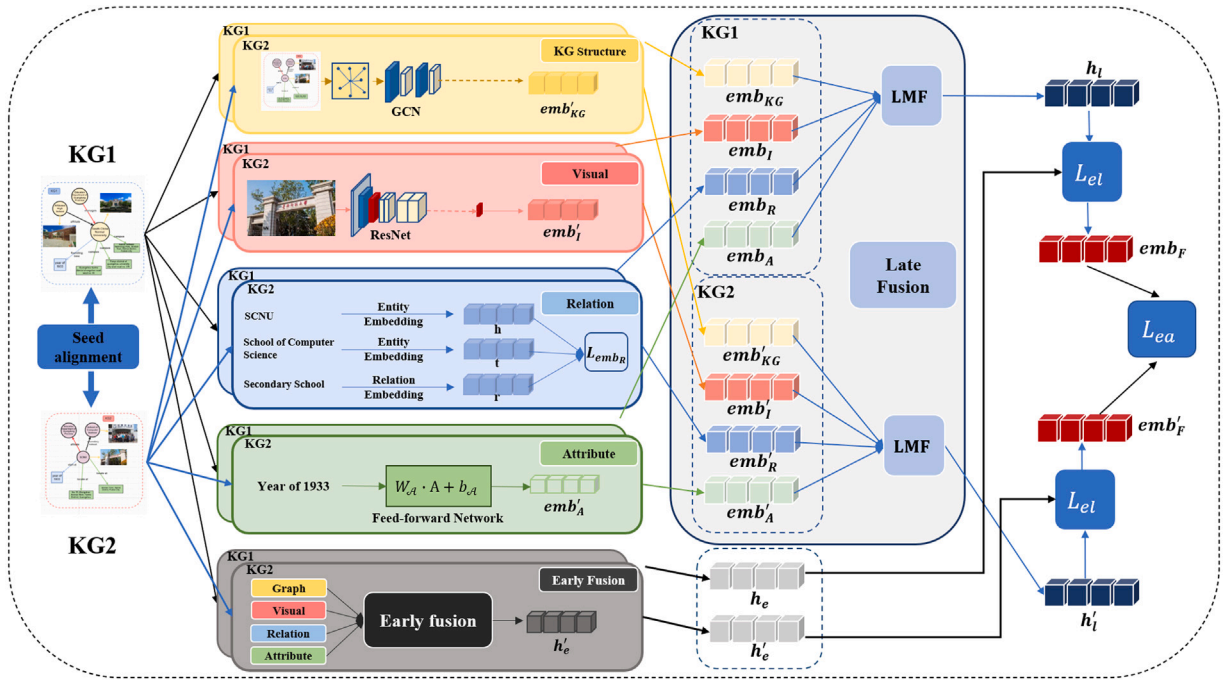


Fig. 2. Overview of DFMKE Framework.

Specifically, we first adopt early fusion to combine various features of multi-modal data. In light of the observations above, an early fusion step aims to exploit correlations between low-level features of each modality. Though this step cannot make full use of the complementary between modalities, it can alleviate the inconsistency between the original data in each modality which is beneficial for later steps.

After that, late fusion is carried out to fuse the early fusion module's output and the multi-modal representation computed via the classical low-rank multi-modal fusion model (LMF), which has been proved efficient to leverage knowledge from multiple modalities [53].

In this section, we will give the technical details about the early and late fusion methods and their combination strategy.

Early Fusion. Before feeding into a model, early fusion is combining feature representations at the input level, which can better capture the relationship among features [54]. We fuse multiple features for various data modalities in this work by using standard early fusion techniques. To combine learned feature representation for images from the final fully-connected layer of the RESNET with features from the other modalities, we create a straightforward fully connected neural network model (NN) as shown in Fig. 3. The early fusion module maps heterogeneous features in a shared space. After that, the output vector h_e of the NN will be combined with the output representation of LMF for late fusion.

Late Fusion. As we mentioned earlier, early fusion can only ensure data consistency between different modalities but cannot guarantee the complementary between data features. Based on these premises, we introduce the late fusion module in the proposed framework, which addresses the complementary problem by fully exploiting the correlations among data features.

Late fusion combines outputs of models for different modalities to mine the complementary [55]. The multi-modal fusion can be formulated as a multi-linear function $f : V_1 \times V_2 \times \dots \times V_M \rightarrow H$, where $V_1, V_2, \dots, V_M$ are the vector spaces of input modalities and H is the output vector space.

Given a set of vector representations, $\{z_m\}_{m=1}^M$, which are encoding uni-modal information of the M different modalities, the goal of multi-modal fusion is to combine the uni-modal representations into one compact multi-modal representation.

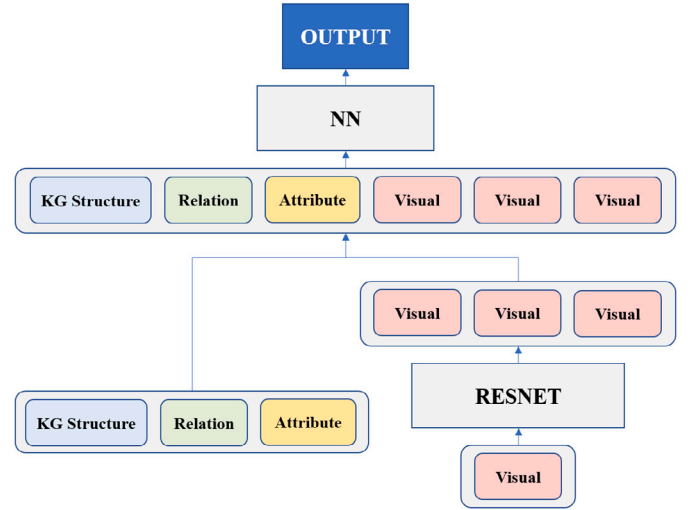


Fig. 3. Early fusion model architecture.

Tensor representation is one of the successful methods for multi-modal fusion. Prior to calculating the outer product, by adding 1s to the uni-modal representations, the input tensor $\mathcal{Z}$ formed by the uni-modal representation is computed as $\mathcal{Z} = \bigotimes_{m=1}^M z_m$, $z_m \in \mathbb{R}^{d_m}$ [56], where $\bigotimes_{m=1}^M$ denotes the tensor outer product over a set of vectors indexed by m . Then, $\mathcal{Z} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_M}$ can be integrated as an order- M tensor with the weight $\mathcal{W}$ using a linear layer as $h = \mathcal{W} \cdot \mathcal{Z} + b$. However, the weight tensor's number of parameters will grow exponentially as well, requiring a lot of computation and putting the model in danger of over-fitting.

By breaking down the weight into a collection of low-rank factors, the LMF solves the computational issues outlined above. Using the fact that $\mathcal{Z} = \bigotimes_{m=1}^M z_m$, the computation for the vector representation h_l can be simplified as:

$$h_l = \bigwedge_{m=1}^M \left[\sum_{i=1}^r \mathbf{w}_m^{(i)} \cdot z_m \right], \quad (8)$$

where $\bigwedge_{m=1}^M$ denotes the element-wise product over a sequence of tensors, r is the rank of the tensor and $\mathbf{w}_m^{(i)}$ is the corresponding low-rank factors for each modality m . Compared to the original computation method in the LMF, where $h_i = \left(\sum_{m=1}^r \bigotimes_{m=1}^M \mathbf{w}_m^{(i)}\right) \cdot \mathcal{Z}$, in the proposed framework, we may compute h_i without actually generating the tensor $\mathcal{Z}$ from the input representations z_m by making use of the parallel decomposition of both $\mathcal{Z}$ and $\mathcal{W}$ in this simplification (Eq. (8)). In doing so, we eliminate the weight-lifting associated with computing the massive input tensors $\mathcal{Z}$ and $\mathcal{W}$.

To achieve the complementarity of heterogeneous data, LMF implicit access to high-dimensional tensors to obtain the data commonalities of different modalities in a high-dimensional space while the unimodal representations are integrated into a compact multi-modal representation.

Moreover, in this work, as we tend to perform joint training with the early fusion module further, the computation is still heavy if r is too big. Therefore, we update Eq. (8) by swapping the order of summation and the element-wise product as below:

$$h_i = \sum_{l=1}^r \left[\bigwedge_{m=1}^M [\mathbf{w}_m^{(1)}, \mathbf{w}_m^{(2)}, \dots, \mathbf{w}_m^{(r)}] \cdot \hat{z}_m \right]_i \quad (9)$$

$$\text{s.t. } r \leq \frac{\prod_{m=1}^M}{\sum_{m=1}^M},$$

where $[\cdot]_i$ indicates the i th slice of a matrix, the Restriction can ensure the decomposition exists in an acceptable range to reduce the computation. In other words, it allows us to early stop the calculation, thus achieving relevant computational savings. This change also allows us to parameterize the model with order- M tensors rather than a collection of vectors, which means we can do the calculation in a parallel fashion in practice is also beneficial from a computational standpoint.

Combination. We then design a strategy to combine h_i with h_e from the early fusion model. This combination strategy participates in the joint training of early fusion and late fusion modules. In this way, multiple modalities benefit from each other by inducing an orthogonal mapping matrix from each embedding space to shared space. Let $\mathbf{H}_e$ and $\mathbf{H}_l$ be the combined embedding matrices obtained by storing vectors h_e and h_l for each entity, respectively. The goal is to maximize the agreement between $\mathbf{H}_e$ and $\mathbf{H}_l$ in a unified embedding space to obtain the final multi-modal embeddings emb_F . The loss function $\mathcal{L}_{el}$ to learn emb_F is designed as follows:

$$\mathcal{L}_{el} = \sum_{i=1}^N \left(\left\| \mathbf{H}_e^i - \mathbf{H}_l^i \mathbf{Z}^{(i)} \right\|_F^2 + \left\| \mathbf{I} - \mathbf{Z}^{(i)\top} \mathbf{Z}^{(i)} \right\|_F^2 \right), \quad (10)$$

where N is the number of entities, $\mathbf{Z}^{(i)}$ serves as the mapping from the i th entity embedding space to the shared space for entity alignment in the next step.

We observe that the loss function $\mathcal{L}_{el}$ is additive, and, in addition, each term appearing in the sum is, itself, the sum of the squares of two norms. Consequently, the function $\mathcal{L}_{el}$ is convex because it is the sum of convex functions. As described earlier, the early fusion module produces a vector (h_e) that ensures data consistency in different modalities, while the late fusion model produces a vector (h_l) that ensures data is complementary across different modalities. With this strategy, the fusion of low-level features and high-dimensional tensors mapped back to a low-dimensional output vector space is implemented, which means the information of modal correlations at the feature level is employed. It is worth observing that Eq. (10) depends only on matrices $\mathbf{H}_e^i$ and $\mathbf{H}_l^i$ for $i \in [1, N]$ and, thus, it does not require to specify in advance seed entities, that is pairs of entities from two KGs which represent the same entity. Consequently, Eq. (10) defines a measure of similarity between entities that can be computed in a fully unsupervised fashion.

From a practical standpoint, we have many efficient options to optimize $\mathcal{L}_{el}$, and to this extent, the stochastic gradient descent method is the most convenient strategy. In the following subsection, we give details on how the training phase was performed.

3.3. Training details

We design a training stage that learns all multi modalities embeddings and iteratively optimizes the shared space learning during an epoch. To regularize embedding vectors, we limit all entity embeddings with L_2 normalization. The parameters are initialized with Xavier initializer, and the loss functions are optimized with Adadelta to simplify the calculation. Every embedding has dimensions set to 100. We use the mini-batch technique with a 1000 batch size. All learning rates were set to 0.01, except shared space learning, which was set to 0.001. In addition, 200 epochs are the maximum allowed.

On top of the emb_F for all entities, we estimate the similarity of all bi-graph entity pairs and align them using a loss function $\mathcal{L}_{ea}$ because the similarity should be high if two entities refer to the same object (the far right of Fig. 2).

Let emb_F^s and emb_F^t denote embeddings of the source and target entities $\mathcal{E}_s$ and $\mathcal{E}_t$ respectively. We compute their cosine similarity matrix $\mathbf{S} = \langle \mathbf{F}_J^s, \mathbf{F}_J^t \rangle \in \mathbb{R}^{|\mathcal{E}_s| \times |\mathcal{E}_t|}$, where the cosine similarity between the i th item in $\mathcal{E}_s$ and the j th entity in $\mathcal{E}_t$ is represented by each entry S_{ij} . The loss function of the whole entity alignment framework $\mathcal{L}_{ea}$ can be formulated as below [57]:

$$\mathcal{L}_{ea} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{\alpha} \log \left(1 + \sum_{m \neq i} e^{\alpha S_{mi}} \right) + \frac{1}{\alpha} \log \left(1 + \sum_{n \neq i} e^{\alpha S_{in}} \right) - \log \left(1 + \beta S_{ii} \right) \right), \quad (11)$$

where α and β are temperature scales; N is the number of seeds. Before obtaining the 4096-dimensional embeddings for each entity, we train RESNET image embeddings. We then train relation embedding from TransE for all triplets. At each training step, the parameters are updated by $\mathcal{L}_{emb_R}$, the loss function of dual fusion scheme $\mathcal{L}_{el}$ and $\mathcal{L}_{ea}$. Once the training process converges, we perform alignment by the nearest-neighbor search based on emb_F . Due to the convexity and additive nature of the functions to be optimized, we employ stochastic gradient descent (SGD) to optimize all of the aforementioned loss functions. In other words, the SGD can converge to the optimal solution using a single iteration to renew the model parameters, significantly accelerating the convergence rate.

4. Experiments

In this part, we assess DFMKE using two real-world datasets and show how it may be used to attain cutting-edge performance by utilizing multi-modal knowledge in the entity alignment task.

4.1. Experimental settings

Datasets. We use two public multi-modal datasets in the experiments, FB15K-DB15K and FB15K-YAGO15K [44]. The FB15K subset was extracted from the Freebase knowledge base as a representative subset. We chose DB15K from DBpedia and YAGO15K from YAGO to keep an approximate entity number aligned with FB15K. The statistics for multi-modal data are shown in Table 1.

Evaluation Metrics. To calculate the similarity between two entities, we use cosine similarity. We use three popular indicators, Hits@n, MR, and MRR, as metrics to evaluate model performance. Hits@n denotes the rate at which correct entities rank in the top n entities based on the similarity computation. The terms MR and MRR in the parameter stand for the mean rank and mean reciprocal rank of correct entities, respectively.

Comparative Methods. We have chosen various types of state-of-the-art models to demonstrate the performance of the DFMKE framework, including two typical translation-based methods, TransE [9], and IPTransE [23]; two straightforward late fusion methods, MMKG [44], and MMEA [10]; and two recently proposed methods, that is MultiKE [12], and EVA [11].

We employ their reported results for those methods using the same datasets. To replicate their experiments, we followed the same hyperparameters setting for other methods mentioned in the original papers.

Table 1
Statistics of multi-modal datasets.

Datasets	Entities	Relations	Attributes	Relational triples
FB15K	14951	1345	116	592213
DB15K	12842	279	225	89197
YAGO15K	15404	32	7	122886

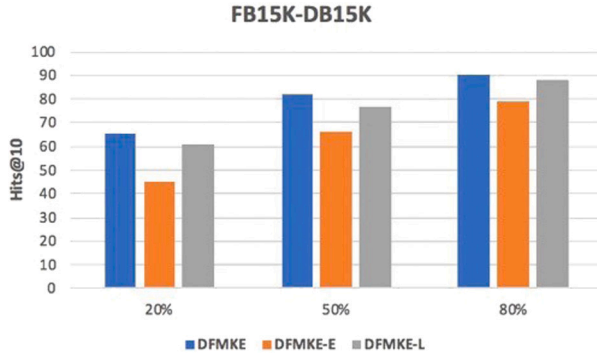


Fig. 4. Ablation study on FB15K-DB15K dataset (Hits@10).

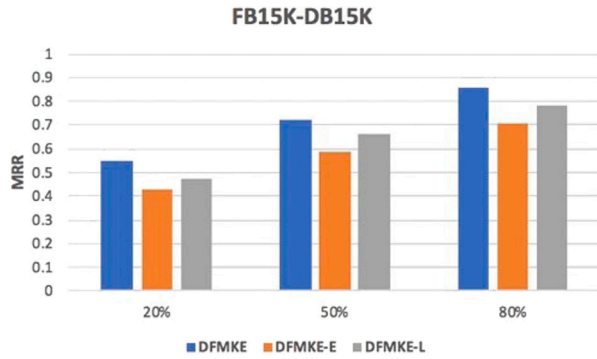


Fig. 5. Ablation study on FB15K-DB15K dataset (MRR).

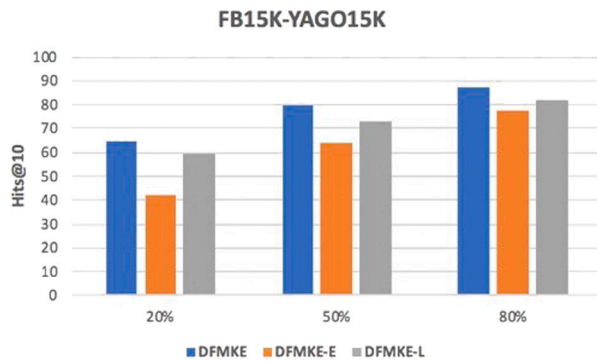


Fig. 6. Ablation study on FB15K-YAGO15K dataset (Hits@10).

4.2. Experimental results

To compare the outcomes of each model, we divide up the datasets. We use 20% of each dataset's data as training sets because this setting can better demonstrate the robustness and effectiveness of all methods, as many existing works prove. The remaining data are treated as testing sets.

Performance Comparison. On the FB15K-DB15K and FB15K-YAGO15K datasets, Table 2 provides all model results with 20% alignment information. In general, the proposed DFMKE outperforms all

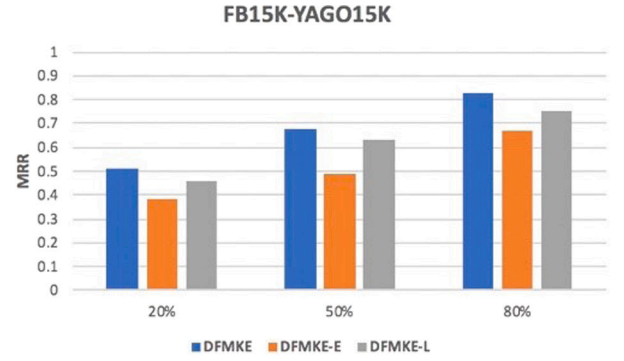


Fig. 7. Ablation study on FB15K-YAGO15K dataset (MRR).

baseline methods on both datasets. For example, DFMKE exceeds 10% more than EVA, which is the second-best method regarding the Hits@n indicator on both datasets. Regarding the MRR indicator, DFMKE is the only model that can exceed 0.5 on both FB15K-DB15K and FB15K-YAGO15K datasets.

Additionally, we found that multi-modal knowledge outperforms those that just use one modality, e.g., TransE and IPTransE, the worst-performing models that use only relation knowledge. The experimental findings demonstrated that DFMKE is more successful for multi-modal knowledge than other fusion approaches.

Analysis of Unsupervised Entity Alignment. This task intends to evaluate the performance of the DFMKE to align entities without seed entities. As found in [58], seed entities are vital to learning KG embeddings from relation facts. However, annotated seeds may not be widely available across KG for entity alignment. Therefore, we considered a variant of our approach – called “DFMKE w/o seed” – in which we assumed no seed entities were available. We compared the DFMKE w/o seed method with the well-known semi-supervised entity alignment method KECG [26] we already introduced in Section 2 on multi-modal KGs. For fairness, we adjusted the architecture of KECG to handle the situation that there are no seed entities.

As shown in Table 3, DFMKE w/o seed still achieves acceptable results, which means the framework does not heavily rely on seed alignment as supervision. At the same time, the relation and attribute views can benefit from early and late fusion during training, even if no seed entity alignment is given. This experiment reveals that DFMKE has good robustness and can alleviate seed alignment reliance. As expected, the Hits@10 and MRR scores achieved by the DFMKE w/o seed approach degrade compared with the DFMKE model in both FB15K-DB15K and FB15K-YAGO15K datasets, but such a worsening appears to be limited. Consequently, we can conclude that lack of seed entities has a negligible impact on the accuracy of the DFMKE approach.

We also point out that DFMKE w/o seed achieves better performance than that observed for KECG in both Hits@10 and MRR metrics; this behavior depends on the fact that the relation and attribute views can benefit from early and late fusion during training, even if no seed entity alignment is given. Therefore, we can conclude that the experiment presented in this section shows that the DFMKE approach displays good robustness and slightly relies on seed entities.

Ablation Study. To further validate the effectiveness of DFMKE, we design two variants for the ablation study, namely DFMKE-E and DFMKE-L. DFMKE-E is a variant of DFMKE using the early fusion method only, and DFMKE-L is a variant of DFMKE using the late fusion method only.

In this experiment, we further evaluate DFMKE, DFMKE-E, and DFMKE-L with different test splits on FB15K-DB15K and FB15K-YAGO15K datasets using Hit@10 and MRR indicators. For instance, 80% means we use the 80% dataset as the training dataset, and the rest of 20% is the test dataset.

Table 2
Results of unsupervised entity alignment.

Models	FB15K-DB15K				FB15K-YAGO15K			
	Hits@1	Hits@10	MR	MRR	Hits@1	Hits@10	MR	MRR
TransE [9]	7.81	24.11	442.47	0.13	6.36	20.25	522.54	0.11
IPTransE [23]	3.99	17.27	387.51	0.08	3.08	14.43	522.23	0.07
MMKG [44]	12.01	25.62	148.82	0.17	10.91	24.14	168.88	0.15
MMEA [10]	26.48	54.11	124.81	0.36	23.39	48.11	147.44	0.32
MultiKE [12]	28.82	56.35	110.85	0.42	28.55	50.25	129.65	0.39
EVA [11]	30.25	59.85	99.65	0.48	32.68	58.65	110.56	0.45
DFMKE	33.82	65.45	89.66	0.55	36.78	64.55	102.53	0.51

Table 3
Results of entity alignment.

	FB15K-DB15K		FB15K-YAGO15K	
	Hits@10	MRR	Hits@10	MRR
KECG [26]	60.11	0.48	58.55	0.46
DFMKE w/o seed	63.21	0.53	62.75	0.49
DFMKE	65.45	0.55	64.55	0.51

Table 4
Ablation study of DFMKE based on 20% alignment data.

	FB15K-DB15K		FB15K-YAGO15K	
	Hits@10	MRR	Hits@10	MRR
w/o structure	40.85	0.41	40.68	0.38
w/o image	62.33	0.52	60.32	0.47
w/o relation	61.45	0.51	59.78	0.46
w/o attribute	61.66	0.52	59.93	0.47

As illustrated in Figs. 4 to 7, DFMKE outperforms DFMKE-E and DFMKE-L, revealing that DFMKE provides insights to take advantage of multi-modal knowledge compared to early or late fusion modules. Moreover, early fusion complements the information absence of feature correlation to improve its overall performance through its performance is much worse than late fusion. Finally, we discover that the performance of all three models on both indicators improved if the amount of training data increased.

In summary, all the above experiments demonstrate that the DFMKE framework can locate entities referencing the same real-world identity from various KGs by fully utilizing multi-modal knowledge to reach cutting-edge performance for entity alignment.

Last but not least, we also perform experiments to verify the impact of each modality on the DFMKE. As shown in Table 4, we have “w/o x” methods in this experiment, where “x” represents the modality we remove. For instance, according to the results, dropping KG structure embedding decreases Hit@10 from 65.45 to 40.85 on FB15K-DB15K, which means the modality remains the most important for our framework. The other three modalities are of equal importance, around a 4%–7% drop for each modality on both datasets. Note that we initialize the same weight for each modality at the beginning of the training. This experiment proves the effectiveness and innovation of our work, especially in the fusion part, as the proposed dual multi-modal knowledge graph embedding framework can leverage knowledge from multiple modalities to achieve better results for entity alignment.

5. Conclusions

In this paper, we propose a dual fusion multi-modal KG embedding framework that integrates several representations of various types of information based on knowledge embedding for the entity alignment task. We first introduce an early fusion method for fusing features of multi-modal entities. Moreover, an efficient late fusion method using modality-specific low-rank factors was designed through shared space learning with the output vectors from early fusion to migrate features under different knowledge spaces. Extensive experiments on two

public datasets showed the robustness and efficiency of the suggested approach for multi-modal entity alignment, which significantly beat state-of-the-art baseline techniques. In the future, we plan to investigate the adaptation of the DFMKE for cross-lingual entity alignment in multi-modal KGs.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhu JIA reports financial support was provided by National Natural Science Foundation of China.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the Key Laboratory of Intelligent Education Technology and Application of Zhejiang Province, Zhejiang Normal University, Zhejiang, China, the Key Research and Development Program of Zhejiang Province (No. 2021C03141), and the National Natural Science Foundation of China under Grant (62077015, 61877020 and 62037001).

References

- [1] D. Wijaya, P.P. Talukdar, T. Mitchell, PIDGIN: ontology alignment using web text as interlingua, in: *ACM International Conference on Information and Knowledge Management*, 2013, pp. 589–598.
- [2] J. Bleiholder, F. Naumann, Data fusion, *ACM Comput. Surv.* 41 (1) (2009) 1–41.
- [3] J. Volz, C. Bizer, M. Gaedke, G. Kobilarov, Discovering and maintaining links on the web of data, in: *Proceedings of International Semantic Web Conference*, 2009, pp. 650–665.
- [4] B.D. Trisedya, J. Qi, R. Zhang, Entity alignment between knowledge graphs using attribute embeddings, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019, pp. 297–304.
- [5] F. Mahdisoltani, J. Biega, F. Suchanek, Yago3: A knowledge base from multilingual wikipedias, in: *7th Biennial Conference on Innovative Data Systems Research*, CIDR Conference, 2014.
- [6] W. Hu, J. Chen, Y. Qu, A self-training approach for resolving object coreference on the semantic web, in: *Proc. of the 20th International Conference on World Wide Web*, Hyderabad, India, 2011, pp. 87–96.
- [7] S.C. Pei, L. Yu, R. Hoehndorf, X.L. Zhang, Semi-supervised entity alignment via knowledge graph embedding with awareness of degree difference, in: *Proceedings of the World Wide Web Conference*, 2019, pp. 3130–3136.
- [8] C. Li, Y. Cao, L. Hou, J. Shi, J. Li, T.-S. Chua, Semi-supervised entity alignment via joint knowledge embedding model and cross-graph model, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, EMNLP-IJCNLP, 2019, pp. 2723–2732.
- [9] A. Bordes, N. Usunier, A. García-Durán, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, in: *Proceedings of Neural Information Processing Systems*, 2013, pp. 2787–2795.
- [10] L. Chen, Z. Li, Y. Wang, T. Xu, Z. Wang, E. Chen, MMEA: Entity alignment for multi-modal knowledge graph, in: *Proceedings of International Conference on Knowledge Science, Engineering and Management*, 2020, pp. 134–146.
- [11] F. Liu, M. Chen, D. Roth, N. Collier, Visual pivoting for (unsupervised) entity alignment, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021, pp. 1–9.

- [12] Q. Zhang, Z. Sun, W. Hu, M. Chen, L. Guo, Y. Qu, Multi-view knowledge graph embedding for entity alignment, in: *Proceedings of the International Joint Conference on Artificial Intelligence*, 2019, pp. 5429–5435.
- [13] G. Klyne, Resource description framework (RDF): Concepts and abstract syntax, 2004, <http://www.w3.org/tr/2004/rec-rdf-concepts-20040210/>.
- [14] A. El-Roby, A. Aboulmaga, ALEX: Automatic link exploration in linked data, in: *Proc. of the 2015 ACM SIGMOD International Conference on Management of Data*, Melbourne, Australia, 2015, pp. 1839–1853.
- [15] Y. Raimond, C. Sutton, M. Sandler, Automatic interlinking of music datasets on the semantic web, in: *Proceedings of LBOw*, 2008, pp. 1–8.
- [16] F.M. Suchanek, S. Abiteboul, P. Senellart, PARIS: probabilistic alignment of relations, instances, and schema, *Proc. VLDB Endow.* 5 (3) (2011) 157–168.
- [17] M. Chen, Y. Tian, M. Yang, C. Zaniolo, Multilingual knowledge graph embeddings for cross-lingual knowledge alignment, in: *Proc. of the Twenty-Sixth International Joint Conference on Artificial Intelligence*, Melbourne, Australia, 2017, pp. 1511–1517.
- [18] Y. Cao, Z. Liu, C. Li, Z. Liu, J. Li, T. Chua, Multi-channel graph neural network for entity alignment, in: *Proc. of the 57th Conference of the Association for Computational Linguistics*, ACL, Association for Computational Linguistics, Florence, Italy, 2019, pp. 1452–1461.
- [19] M. Chen, Y. Tian, K. Chang, S. Skiena, C. Zaniolo, Co-training embeddings of knowledge graphs and entity descriptions for cross-lingual entity alignment, in: *Proc. of the Twenty-Seventh International Joint Conference on Artificial Intelligence*, IJCAI 2018, Stockholm, Sweden, 2018, pp. 3998–4004.
- [20] S. Pei, L. Yu, R. Hoehndorf, X. Zhang, Semi-supervised entity alignment via knowledge graph embedding with awareness of degree difference, in: *The World Wide Web Conference*, WWW 2019, San Francisco, CA, USA, May 13–17, 2019, ACM, 2019, pp. 3130–3136.
- [21] Z. Sun, W. Hu, Q. Zhang, Y. Qu, Bootstrapping entity alignment with knowledge graph embedding, in: *Proceedings of the International Joint Conference on Artificial Intelligence*, 2018, pp. 4396–4402.
- [22] Y. Wu, X. Liu, Y. Feng, Z. Wang, R. Yan, D. Zhao, Relation-aware entity alignment for heterogeneous knowledge graphs, in: *Proc. of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, IJCAI 2019, Macao, China, pp. 5278–5284.
- [23] H. Zhu, R. Xie, Z. Liu, M. Sun, Iterative entity alignment via joint knowledge embeddings, in: *Proceedings of the International Joint Conference on Artificial Intelligence*, 2017, pp. 4258–4264.
- [24] C. Li, Y. Cao, L. Hou, J. Shi, J. Li, T.-S. Chua, Semi-supervised entity alignment via knowledge graph embedding with awareness of degree difference, in: *Proceedings of the World Wide Web Conference*, 2019, pp. 3130–3136.
- [25] S. Pei, L. Yu, R. Hoehndorf, X. Zhang, Semi-supervised entity alignment via knowledge graph embedding with awareness of degree difference, in: *The World Wide Web Conference*, 2019, pp. 3130–3136.
- [26] C. Li, Y. Cao, L. Hou, J. Shi, J. Li, T.-S. Chua, Semi-supervised entity alignment via joint knowledge embedding model and cross-graph model, in: *EMNLP-IJCNLP*, 2019, pp. 2723–2732.
- [27] Z. Wang, J. Zhang, J. Feng, Z. Chen, Knowledge graph embedding by translating on hyperplanes, in: *Proc. of the Twenty-Eighth AAAI Conference on Artificial Intelligence*, AAAI Press, Québec City, Québec, Canada, 2014, pp. 1112–1119.
- [28] Y. Lin, Z. Liu, M. Sun, Y. Liu, X. Zhu, Learning entity and relation embeddings for knowledge graph completion, in: *Proc. of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, January, AAAI Press, Austin, Texas, USA, 2015, pp. 2181–2187.
- [29] B. Yang, W. Yih, X. He, J. Gao, L. Deng, Embedding entities and relations for learning and inference in knowledge bases, in: *Proc. of the International Conference on Learning Representations*, ICLR 2015, San Diego, CA, USA, 2015.
- [30] M. Nickel, L. Rosasco, T.A. Poggio, Holographic embeddings of knowledge graphs, in: *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*, February 12–17, 2016, Phoenix, Arizona, USA, AAAI Press, 2016, pp. 1955–1961.
- [31] Z. Sun, Z. Deng, J. Nie, J. Tang, Rotate: Knowledge graph embedding by relational rotation in complex space, in: *7th International Conference on Learning Representations*, ICLR 2019, New Orleans, LA, USA, 2019.
- [32] B. Shi, T. Weninger, Proje: Embedding projection for knowledge graph completion, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2017.
- [33] T. Dettmers, P. Minervini, P. Stenetorp, S. Riedel, Convolutional 2D knowledge graph embeddings, in: *Proc. of the Thirty-Second AAAI Conference on Artificial Intelligence*, AAAI-18, AAAI Press, New Orleans, Louisiana, USA, 2018, pp. 1811–1818.
- [34] Q. Wang, Z. Mao, B. Wang, L. Guo, Knowledge graph embedding: A survey of approaches and applications, *IEEE Trans. Knowl. Data Eng.* 29 (12) (2017) 2724–2743.
- [35] A. Holzinger, B. Malle, A. Saranti, B. Pfeifer, Towards multi-modal causability with graph neural networks enabling information fusion for explainable AI, *Inf. Fusion* 71 (2021) 28–37.
- [36] R. Krishna, Y. Zhu, O. Groth, J. Johnson, K. Hata, J. Kravitz, S. Chen, Y. Kalantidis, L.-J. Li, D.A. Shamma, et al., Visual genome: Connecting language and vision using crowdsourced dense image annotations, *Int. J. Comput. Vis.* 123 (1) (2017) 32–73.
- [37] D. Oñoro-Rubio, M. Niepert, A. García-Durán, R. González-Sánchez, R.J. López-Sastre, Representation learning for visual-relational knowledge graphs, 2017, *arXiv preprint arXiv:1709.02314*.
- [38] R. Xie, Z. Liu, H. Luan, M. Sun, Image-embodied knowledge representation learning, in: *Proceedings of International Joint Conference on Artificial Intelligence*, 2017, pp. 3140–3146.
- [39] P. Pezeshkpoor, L. Chen, S. Singh, Embedding multimodal relational data for knowledge base completion, in: *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 3208–3218.
- [40] A. Krizhevsky, I. Sutskever, G.E. Hinton, ImageNet classification with deep convolutional neural networks, in: *Proc. of the International Conference on Neural Information Processing Systems*, NIPS 2012, Lake Tahoe, Nevada, United States, 2012, pp. 1106–1114.
- [41] P. Verga, D. Belanger, E. Strubell, B. Roth, A. McCallum, Multilingual relation extraction using compositional universal schema, in: *Proc. of TheConference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, The Association for Computational Linguistics, San Diego, California, USA, 2016, pp. 886–896.
- [42] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, in: *Proc. of the International Conference on Learning Representations*, ICLR 2015, San Diego, California, USA, 2015.
- [43] Y. Gao, O. Beijbom, N. Zhang, T. Darrell, Compact bilinear pooling, in: *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 317–326.
- [44] Y. Liu, H. Li, A. Garcia-Duran, M. Niepert, D. Onoro-Rubio, D.S. Rosenblum, MMKG: multi-modal knowledge graphs, in: *Proceedings of the European Semantic Web Conference*, 2019, pp. 459–474.
- [45] A. García-Durán, M. Niepert, KBIR: End-to-end learning of knowledge base representations with latent, relational, and numerical features, in: *Proc. of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence*, UAI 2018, Monterey, California, USA, 2018, pp. 372–381.
- [46] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, 2017, pp. 1–14.
- [47] J. Zhu, Z. Zheng, M. Yang, G.P.C. Fung, Y. Tang, A semi-supervised model for knowledge graph embedding, *Data Min. Knowl. Discov.* 34 (2019) 1–20.
- [48] B. Perozzi, R. Al-Rfou, S. Skiena, Deepwalk: Online learning of social representations, in: *Proceedings of ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2014, pp. 701–710.
- [49] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [50] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, in: *Proceedings of International Conference on Learning Representations*, 2015, pp. 1–14.
- [51] H.-W. Yang, Y. Zou, P. Shi, W. Lu, J. Lin, X. Sun, Aligning cross-lingual entities with multi-aspect information, in: *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the International Joint Conference on Natural Language Processing*, 2019, pp. 4422–4432.
- [52] P. Atrey, M. Hossain, A.E. Saddik, M.S. Kankanhalli, Multimodal fusion for multimedia analysis: a survey, *Multimedia Syst.* 16 (6) (2010) 345–379.
- [53] Z. Liu, Y. Shen, V.B. Lakshminarasimhan, P.P. Liang, A.B. Zadeh, L.P. Morency, Efficient low-rank multimodal fusion with modality-specific factors, in: *Proceedings of Annual Meeting of the Association for Computational Linguistics*, 2018, pp. 2247–2256.
- [54] Z. Zhong Lan, L. Bao, S.-I. Yu, W. Liu, A.G. Hauptmann, Multimedia classification and event detection using double fusion, *Multimedia Tools Appl.* 71 (1) (2014) 333–347.
- [55] C.G.M. Snoek, M. Worring, A.W.M. Smeulders, Early versus late fusion in semantic video analysis, in: *ACM International Conference Multimedia*, 2005, pp. 399–402.
- [56] A. Zadeh, M. Chen, S. Poria, E. Cambria, L.-P. Morency, Tensor fusion network for multimodal sentiment analysis, in: *Proceedings of Empirical Methods in Natural Language Processing*, 2017, pp. 1103–1114.
- [57] F. Liu, R. Ye, X. Wang, S. Li, HAL: Improved TextImage matching by mitigating visual semantic hubs, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020, pp. 11563–11571.
- [58] Z. Wang, Q. Lv, X. Lan, Y. Zhang, Cross-lingual knowledge graph alignment via graph convolutional networks, in: *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 349–357.